

BUSSELTON MARGARET RIVER, AustraliaWMO: **956110**Lat: **33.6858S**Lon: **115.4008E**Elev: **55**StdP: **14.67**Time Zone: **8.00 (E08)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.0	39.9	34.1	28.8	48.2	36.2	31.4	47.4	27.3	59.6	25.0	59.6	4.3	100	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.0	93.3	66.9	90.0	66.3	86.9	65.4	70.2	84.6	68.7	82.8	67.4	81.0	11.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	97.7	72.4	64.5	91.2	71.5	63.0	86.4	71.0	34.1	85.0	32.9	82.7	31.9	81.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	21.2	19.5	DB	34.3	99.5	1.3	1.9	33.4	100.9	32.7	102.0	32.0	103.1	31.1	104.5	(4)
(5)			WB	33.8	73.3	1.3	1.7	32.8	74.6	32.1	75.6	31.4	76.5	30.5	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	71.1	71.3	68.4	63.0	58.0	54.6	52.9	53.6	54.7	58.1	63.6	67.7
	DBStd	8.25	5.54	5.73	5.75	5.03	3.96	3.68	3.64	3.57	3.86	4.65	5.59	6.20
	HDD50	33	0	0	0	0	1	4	13	8	6	1	0	0
	HDD65	2077	11	10	31	98	220	313	375	355	311	220	92	41
	CDD50	4178	655	597	571	389	248	141	103	118	146	252	409	549
	CDD65	747	201	187	137	37	3	0	0	0	0	7	51	124
	CDH74	7601	2089	1917	1283	265	14	0	0	0	5	86	557	1385
	CDH80	2934	887	805	441	42	0	0	0	0	0	13	174	572
Wind	WSAvg	9.8	12.2	11.9	10.8	8.6	7.9	8.5	8.6	8.6	9.0	9.4	10.5	11.4
Precipitation	PrecAvg	31.1	0.3	0.4	0.8	1.7	4.4	6.5	6.5	4.6	3.0	1.7	1.0	0.5
	PrecMax	45.2	2.8	2.5	3.9	6.4	9.6	12.2	13.0	9.1	5.8	4.8	3.2	2.8
	PrecMin	20.9	0.0	0.0	0.0	0.0	1.3	2.5	2.5	1.4	1.2	0.0	0.0	0.0
	PrecStd	5.2	0.6	0.6	1.0	1.3	2.1	2.4	2.3	1.7	1.2	1.0	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.5	96.0	94.2	84.6	75.9	68.5	66.2	67.6	73.0	81.9	90.1	95.9
		MCWB	67.7	69.0	67.6	63.7	60.4	58.8	59.9	59.8	58.8	62.9	63.8	66.9
	2%	DB	92.7	92.1	88.3	80.3	71.8	66.2	64.2	65.0	68.3	76.1	85.3	91.4
		MCWB	66.9	67.5	65.7	62.8	60.2	59.3	58.3	58.1	58.9	61.1	63.1	66.3
	5%	DB	89.1	88.6	84.7	76.8	69.4	64.6	62.6	63.3	65.6	72.0	80.5	87.0
		MCWB	66.4	66.7	64.7	62.4	60.5	58.2	57.1	57.0	57.8	59.9	62.3	64.8
	10%	DB	85.0	85.3	81.2	73.8	67.5	63.1	61.2	61.7	63.4	68.6	76.2	81.9
		MCWB	65.6	65.6	64.1	61.8	60.0	57.3	56.0	55.9	56.7	58.6	61.3	63.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	73.4	71.0	68.4	66.6	63.6	62.7	62.5	63.4	65.5	67.4	70.0
		MCDB	86.2	85.1	83.9	73.8	68.9	65.4	64.2	64.8	66.9	75.2	80.3	87.7
	2%	WB	70.0	70.0	68.8	66.6	64.5	61.3	60.3	60.3	60.9	63.3	65.7	67.9
		MCDB	85.2	85.1	80.8	72.9	68.3	64.2	62.7	63.4	65.9	72.1	77.9	85.5
	5%	WB	68.1	68.4	67.4	65.0	62.5	59.7	58.4	58.4	58.9	61.4	64.1	66.2
		MCDB	84.1	83.9	78.9	72.7	67.1	63.1	61.4	62.1	64.4	69.4	76.6	82.7
	10%	WB	66.3	66.9	65.9	63.3	60.8	58.1	56.6	56.6	57.3	59.6	62.5	64.5
		MCDB	81.9	81.9	76.9	71.6	66.1	61.9	60.2	60.7	62.5	67.0	74.5	79.3
Mean Daily Temperature Range	5% DB	MDBR	25.5	25.0	23.2	20.8	18.7	16.5	15.9	16.1	17.3	20.3	22.5	24.6
		MCDBR	31.2	29.5	27.4	24.4	21.2	17.8	16.5	17.5	21.0	25.9	29.0	31.4
		MCWBR	11.6	10.6	10.3	11.1	12.7	12.3	11.9	12.1	13.2	13.4	12.2	12.0
	5% WB	MCDBR	26.5	25.3	21.6	18.9	17.1	15.0	14.4	15.1	18.3	21.1	24.6	27.8
		MCWBR	11.2	10.5	9.8	11.0	12.7	12.3	12.2	12.1	13.0	12.6	12.1	11.9
Clear-Sky Solar Irradiance	taub	0.333	0.333	0.322	0.316	0.311	0.309	0.303	0.314	0.329	0.327	0.333	0.327	
	taud	2.591	2.600	2.631	2.641	2.624	2.610	2.627	2.585	2.519	2.543	2.548	2.588	
	Ebn at Noon	318	312	305	289	274	266	274	285	297	311	317	322	
	Edh at Noon	33	32	29	25	23	22	22	26	31	33	34	33	
All-Sky Solar Radiation	RadAvg	2487	2214	1774	1261	912	733	785	1048	1408	1869	2255	2485	
	RadStd	128	91	91	75	62	46	40	62	80	98	131	124	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page